

Ruin, Survival, and Arbitrarily High Finite Targets in a $+2/-1$ Gambling Game

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Abstract

In each round, a gambler either wins two units or loses one unit. The game ends when the capital first reaches zero. Starting with one unit, what is the probability of reaching a very large but finite target? For independent fair rounds, this probability tends to $(3 - \sqrt{5})/2$, not to zero. The ruin probability is the inverse golden ratio. We first use path counting to show that ruin is not certain. The strong Markov property then gives $q_{x+y} = q_x q_y$, so the ruin probability from x units is $((\sqrt{5} - 1)/2)^x$.

For a general win probability p , the critical value is $p = 1/3$. We then keep the stationary win rate at $1/2$, but allow the outcomes to follow a symmetric two-state Markov chain. Let θ be the probability that the next outcome repeats the previous one. The survival probability is

$$1 - \frac{1 + r + r^2}{2(1 + r)} r^{x-1}, \quad r = \frac{\sqrt{(1 - \theta)^2 + 4\theta^2} - (1 - \theta)}{2\theta}.$$

It decreases strictly with θ , although the stationary win rate and the mean increment do not change. As $\theta \uparrow 1$, the survival probability tends to $1/4$, but at $\theta = 1$ it is $1/2$. Thus the target limit and the persistence limit do not commute.

Keywords: skip-free random walk; Markov additive process; correlated random walk; first passage; persistence; noncommuting limits.

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1 The problem

Suppose a gambler starts the game with one unit. A fair coin is tossed in each round. A win adds two units and a loss removes one unit. The game ends when the capital reaches zero. The mean increment is

$$\frac{1}{2}(2) + \frac{1}{2}(-1) = \frac{1}{2},$$

but an immediate loss still ruins the gambler. It may seem unreasonable to ask whether one unit can become a million or a billion. The target is very far away. However, the calculation below shows that the main danger is near zero, especially in the first few rounds.

We use three related but different events. Ruin means visiting zero at some finite time, survival means never visiting zero, and crossing m means reaching or exceeding m before ruin. The last wording matters because an upward jump can skip a level. It also explains why the probabilities of crossing $1, 2, 3, \dots$ are not a probability mass function: the same path may cross all of them.

The independent model is a classical skip-free-to-the-left random walk. Such walks have been studied in ruin theory, queueing, and branching processes; see Brown, Peköz, and Ross [2]

and Avram and Vidmar [1]. Correlated gambler's ruin is also classical, beginning at least with Mohan [6]; subsequent treatments include Proudfoot and Lampard [9], Mukherjea and Steele [7], Lal and Bhat [5], and the recent martingale treatment of Pozdnyakov [8]. General boundary crossing for Markov random walks is treated, for example, by Fuh [4].

The general framework is classical. This paper studies what happens for the particular payoff pair $\{+2, -1\}$. In the independent model, the ruin probability is related to the golden ratio, and the limiting probability of reaching a very high finite target is positive. In the Markov model, an exact formula shows that the order of wins and losses matters even when the stationary win rate and the mean payoff stay fixed. At full persistence there is also a discontinuity, and two limits do not commute. No claim is made that the general theory of ruin or correlated random walks is new.

2 Why one-unit ruin equals the inverse golden ratio

Let

$$q_x = \mathbb{P}_x(\tau_0 < \infty), \quad q_0 = 1,$$

where τ_0 is the first time the capital reaches zero. We first prove that the ruin probability is multiplicative. Starting from $x + y$ units, every path to zero must first visit y , because each downward move is only one unit. The probability of reaching y from $x + y$ is the same as the probability of reaching zero from x . After the process reaches y , the strong Markov property gives

$$q_{x+y} = q_x q_y, \quad x, y \in \mathbb{Z}_{\geq 0}. \quad (2.1)$$

In particular,

$$q_x = q_1^x, \quad q_3 = q_1^3. \quad (2.2)$$

First-step analysis from one unit gives

$$q_1 = \frac{1}{2}q_0 + \frac{1}{2}q_3 = \frac{1}{2} + \frac{1}{2}q_1^3. \quad (2.3)$$

Hence

$$(q_1 - 1)(q_1^2 + q_1 - 1) = 0,$$

so the two possible probability roots are

$$q_1 = 1 \quad \text{or} \quad q_1 = \alpha := \frac{\sqrt{5} - 1}{2}. \quad (2.4)$$

Equation (2.3) alone does not decide which root is correct. We now rule out $q_1 = 1$ by counting ruin paths. This does not assume the golden ratio in advance.

Suppose a ruin path from one unit contains k wins. Since every win adds two units and the last step reaches zero, the path must contain $2k + 1$ losses and have length $3k + 1$. It must stay positive before the last step. By the cycle lemma (equivalently, the Raney lemma; see Feller [3]), the number of such valid paths is

$$A_k = \frac{1}{3k + 1} \binom{3k + 1}{k} = \frac{1}{2k + 1} \binom{3k}{k}. \quad (2.5)$$

Among the $3k + 1$ cyclic rotations of a word containing k symbols $+2$ and $2k + 1$ symbols -1 , exactly one has nonnegative proper partial sums and total sum -1 . Thus the probability of ruin after exactly k wins is

$$b_k = \frac{1}{2k + 1} \binom{3k}{k} 2^{-(3k+1)},$$

and the disjoint ruin paths give

$$q_1 = \sum_{k=0}^{\infty} b_k. \quad (2.6)$$

We do not need to sum this series. Its first terms are

$$b_0 = \frac{1}{2}, \quad b_1 = \frac{1}{16}, \quad b_2 = \frac{3}{128},$$

and

$$\frac{b_{k+1}}{b_k} = \frac{3(3k+1)(3k+2)}{16(k+1)(2k+3)} < \frac{27}{32}. \quad (2.7)$$

The last inequality is equivalent to $27k + 23 > 0$. Therefore

$$q_1 < \frac{1}{2} + \frac{1}{16} + \frac{3/128}{1 - 27/32} = \frac{57}{80} < 1. \quad (2.8)$$

This estimate rules out the root 1. Therefore,

$$\boxed{q_1 = \alpha = \frac{\sqrt{5} - 1}{2}}, \quad \boxed{q_x = \alpha^x}. \quad (2.9)$$

Thus the golden ratio is obtained at the end of the argument. It is not inserted at the beginning.

3 More capital and arbitrarily high finite targets

The formula also shows what happens when the gambler starts with more money. Each extra unit multiplies the ruin probability by α . Therefore the survival probability approaches one exponentially fast.

Initial capital x	Ruin α^x	Survival $1 - \alpha^x$	Survival (percent)
1	0.618034	0.381966	38.1966%
2	0.381966	0.618034	61.8034%
3	0.236068	0.763932	76.3932%
5	0.090170	0.909830	90.9830%
10	0.008131	0.991869	99.1869%
20	0.000066107	0.999933893	99.9933893%

We now return to the very large target in the opening question. Let $S_0 = x \in \mathbb{Z}_{>0}$ and let every increment take one of the values $+2$ and -1 . Define

$$\tau_0 = \inf\{n \geq 1 : S_n = 0\}, \quad T_m = \inf\{n \geq 0 : S_n \geq m\}, \quad m > x.$$

The walk is stopped at τ_0 . Since downward motion is by one unit, a path cannot jump across zero without visiting it.

Proposition 3.1 (Extreme-target principle). *Assume that, on $\{\tau_0 = \infty\}$,*

$$S_n \longrightarrow +\infty \quad \text{almost surely.}$$

Then

$$\lim_{m \rightarrow \infty} \mathbb{P}_x(T_m < \tau_0) = \mathbb{P}_x(\tau_0 = \infty). \quad (3.1)$$

In particular, conditional on eternal survival, every finite target is crossed almost surely.

Proof. Put $A_m = \{T_m < \tau_0\}$. The events A_m decrease with m . If ruin occurs, the stopped path has finite length and hence a finite maximum, so it does not belong to $\cap_m A_m$. If ruin never occurs, the assumed divergence of S_n implies membership in every A_m . Therefore

$$\bigcap_{m>x} A_m = \{\tau_0 = \infty\}$$

up to a null set. Continuity of probability from above gives (3.1). $\square$

This result concerns eventual success. It does not say how long success will take. The waiting time has the following asymptotic law.

Proposition 3.2 (Time cost of a high target). *Suppose that, on eternal survival,*

$$\frac{S_n}{n} \longrightarrow \mu > 0 \quad \text{almost surely.}$$

Then, on $\{\tau_0 = \infty\}$,

$$\frac{T_m}{m} \longrightarrow \frac{1}{\mu} \quad \text{almost surely.} \quad (3.2)$$

Proof. For every $\varepsilon \in (0, \mu)$, we eventually have $(\mu - \varepsilon)n \leq S_n \leq (\mu + \varepsilon)n$. At the first crossing, $m \leq S_{T_m} \leq m + 1$, because the upward jump has size two. The usual inverse bounds therefore give $(\mu + \varepsilon)^{-1} \leq \liminf T_m/m \leq \limsup T_m/m \leq (\mu - \varepsilon)^{-1}$. Let $\varepsilon \downarrow 0$. $\square$

In the fair game, the strong law gives $S_n/n \rightarrow 1/2$ on every unrestricted sample path, and hence on the survival event. Propositions 3.1 and 3.2, together with (2.9), give

$$\lim_{m \rightarrow \infty} \mathbb{P}_x(T_m < \tau_0) = 1 - \alpha^x, \quad \frac{T_m}{m} \longrightarrow 2 \quad \text{on survival.} \quad (3.3)$$

Thus a gambler starting with twenty units has a probability of at least 99.9933893% of crossing any given finite target, however large; the exact finite-target probability is slightly larger than the limit. The infinite time horizon is important here. On a surviving path, crossing one hundred million units takes roughly two hundred million rounds. A probability near one may still come with a very long waiting time.

4 Changing the win probability and the critical point

Let the increments be independent, with

$$\mathbb{P}(X_n = 2) = p, \quad \mathbb{P}(X_n = -1) = q = 1 - p, \quad S_n = x + \sum_{j=1}^n X_j.$$

The mean increment is $3p - 1$. Therefore $p_c = 1/3$ is the critical value for the drift.

Theorem 4.1 (Ultimate ruin for independent trials). *For every $x \in \mathbb{Z}_{>0}$:*

- (i) *if $p \leq 1/3$, then $\mathbb{P}_x(\tau_0 < \infty) = 1$;*
- (ii) *if $p > 1/3$, there is a unique $\rho_p \in (0, 1)$ satisfying*

$$p\rho_p^3 - \rho_p + q = 0,$$

and

$$\mathbb{P}_x(\tau_0 < \infty) = \rho_p^x, \quad \rho_p = \frac{\sqrt{4/p - 3} - 1}{2}. \quad (4.1)$$

Consequently,

$$\lim_{m \rightarrow \infty} \mathbb{P}_x(T_m < \tau_0) = \begin{cases} 0, & p \leq 1/3, \\ 1 - \rho_p^x, & p > 1/3. \end{cases} \quad (4.2)$$

Proof. Let u_x be the ultimate ruin probability. First-step analysis gives

$$u_x = pu_{x+2} + qu_{x-1}, \quad u_0 = 1.$$

Substituting $u_x = r^x$ gives

$$pr^3 - r + q = (r-1)(pr^2 + pr - q) = 0.$$

For $p > 1/3$, the quadratic factor has the root $\rho_p \in (0, 1)$ shown in (4.1). Then $\rho_p^{S_n \wedge \tau_0}$ is a bounded martingale. On ruin it converges to one. On survival, the strong law gives $S_n/n \rightarrow 3p - 1 > 0$, so the martingale converges to zero. Bounded convergence proves (4.1). If $p < 1/3$, the unrestricted walk tends to $-\infty$. If $p = 1/3$, the nondegenerate zero-mean finite-variance walk oscillates. In both cases, downward skip-freeness implies that the walk visits zero almost surely. Equation (4.2) follows from Proposition 3.1. $\square$

For $p = 1/2$, formula (4.1) gives the inverse golden ratio already obtained by path counting. At the critical value, the limiting target probability is zero, but the decay is of order $1/m$, not exponential.

Theorem 4.2 (Exact critical high-target law). *Let $p = 1/3$. For integers $1 \leq x < m$,*

$$\mathbb{P}_x(T_m < \tau_0) = \frac{3x(-2)^m + (-2)^x - 1}{(3m+1)(-2)^m - 1}. \quad (4.3)$$

For fixed x ,

$$\mathbb{P}_x(T_m < \tau_0) \sim \frac{x}{m} \quad (m \rightarrow \infty).$$

Proof. Write $h_x = \mathbb{P}_x(T_m < \tau_0)$. Then

$$h_x = \frac{1}{3}h_{x+2} + \frac{2}{3}h_{x-1}, \quad h_0 = 0, \quad h_m = h_{m+1} = 1.$$

The characteristic polynomial is

$$r^3 - 3r + 2 = (r-1)^2(r+2),$$

so $h_x = A + Bx + C(-2)^x$. Imposing the three boundary values gives (4.3). Dividing numerator and denominator by $(-2)^m$ gives the asymptotic formula. $\square$

For example, starting from one unit, the exact probabilities of crossing targets 4, 5, 10 are $5/23$, $11/57$, and $341/3527$, respectively. These values also show that the asymptotic value $1/m$ is not an exact formula for a finite target.

5 The same win rate, but Markov-dependent outcomes

So far, the outcomes have been independent. We now allow dependence without changing the win rate in one round. Let $J_n \in \{W, L\}$ be a stationary Markov chain with transition matrix

$$P_\theta = \begin{pmatrix} \theta & 1-\theta \\ 1-\theta & \theta \end{pmatrix}, \quad 0 < \theta < 1, \quad (5.1)$$

and initial distribution $(1/2, 1/2)$. A win contributes $+2$ and a loss contributes -1 . The parameter θ is the probability that the next outcome repeats the current outcome.

For every θ , the stationary win probability is $1/2$, and the mean increment is also $1/2$. Only the order of the outcomes changes. A large θ produces longer runs of the same outcome. If $Y_n = 2\mathbf{1}_{\{J_n=W\}} - \mathbf{1}_{\{J_n=L\}}$, then

$$\text{Cov}(Y_0, Y_k) = \frac{9}{4}(2\theta - 1)^k, \quad (5.2)$$

and, for $0 < \theta < 1$,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \text{Var} \left(\sum_{j=1}^n Y_j \right) = \frac{9}{4} \frac{\theta}{1 - \theta}. \quad (5.3)$$

Thus the one-round distribution alone does not show the increase in long-run variance.

Theorem 5.1 (Exact Markov ruin law). *For $0 < \theta < 1$, let $r = r(\theta)$ be the root in $(0, 1)$ of*

$$\theta r^2 + (1 - \theta)r - \theta = 0, \quad (5.4)$$

namely

$$r(\theta) = \frac{\sqrt{(1 - \theta)^2 + 4\theta^2} - (1 - \theta)}{2\theta}. \quad (5.5)$$

If the first outcome is sampled from the stationary distribution, then the ultimate ruin and survival probabilities from $x \geq 1$ are

$$\mathbb{P}_x(\tau_0 < \infty) = C(r)r^{x-1}, \quad \mathbb{P}_x(\tau_0 = \infty) = 1 - C(r)r^{x-1}, \quad (5.6)$$

where

$$C(r) = \frac{1 + r + r^2}{2(1 + r)}. \quad (5.7)$$

The same quantity is the extreme-target limit:

$$\lim_{m \rightarrow \infty} \mathbb{P}_x(T_m < \tau_0) = 1 - C(r)r^{x-1}. \quad (5.8)$$

Moreover, on survival, $T_m/m \rightarrow 2$ almost surely.

Proof. Condition on the outcome immediately preceding the next move. Define

$$f_W(x) = \frac{r^{x-1}}{1 + r}, \quad f_L(x) = r^x, \quad x \geq 1,$$

and set $f_L(0) = 1$. Equation (5.4) gives the two harmonic identities

$$f_W(x) = \theta f_W(x + 2) + (1 - \theta)f_L(x - 1),$$

$$f_L(x) = (1 - \theta)f_W(x + 2) + \theta f_L(x - 1).$$

Therefore, $f_{J_n \wedge \tau_0}(S_{n \wedge \tau_0})$ is a bounded martingale. By the ergodic theorem applied to J_n , $S_n/n \rightarrow 1/2$ almost surely. On survival the martingale therefore converges to zero; on ruin it equals $f_L(0) = 1$. Bounded convergence shows that f_W and f_L are the ruin probabilities conditional on the current outcome state.

Before the first outcome, a win and a loss each have probability $1/2$. Thus, for every $x \geq 1$ (including $x = 1$, where the first loss is immediate ruin),

$$\begin{aligned} \mathbb{P}_x(\tau_0 < \infty) &= \frac{1}{2}f_W(x + 2) + \frac{1}{2}f_L(x - 1) \\ &= \frac{1}{2}r^{x-1} \left(1 + \frac{r^2}{1 + r} \right) = C(r)r^{x-1}. \end{aligned}$$

Propositions 3.1 and 3.2 finish the proof. □

6 Longer winning and losing streaks reduce survival

The table and curve below show the effect of longer runs on survival.

Corollary 6.1 (Persistence penalty). *For every fixed $x \geq 1$, the survival probability in (5.6) is strictly decreasing in $\theta \in (0, 1)$.*

Proof. Solving (5.4) for θ gives

$$\theta = \frac{r}{1 + r - r^2},$$

whose derivative with respect to $r \in (0, 1)$ is $(1 + r^2)/(1 + r - r^2)^2 > 0$. Hence r increases strictly with θ . The function $C(r)$ also increases strictly, and so does $C(r)r^{x-1}$. The survival probability is its complement. $\square$

For one unit of initial capital, the effect is clear:

θ	outcome pattern	extreme-target limit
0.10	strongly alternating	0.494571
0.25	alternating tendency	0.464816
0.50	independent	0.381966
0.75	persistent streaks	0.305746
0.90	strong streaks	0.270068
0.99	very long streaks	0.251888

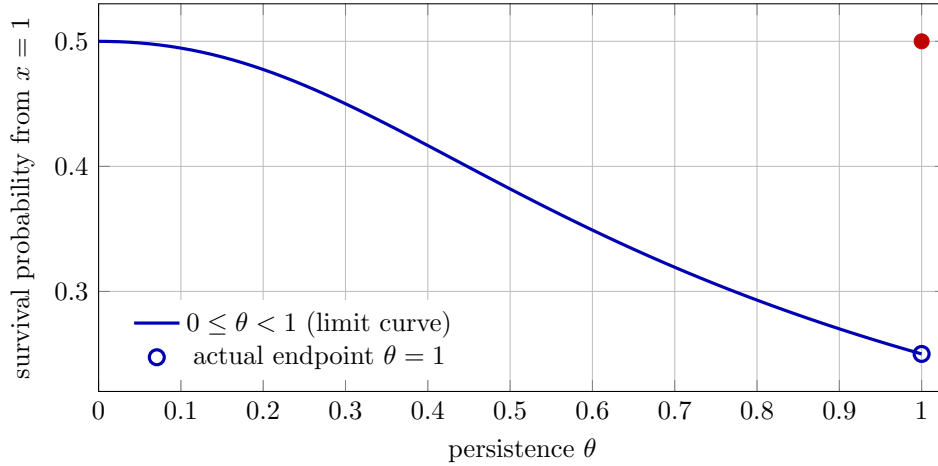


Figure 1: All models have stationary win probability $1/2$ and mean increment $1/2$. Survival decreases with persistence on $[0, 1)$, but the fully frozen endpoint jumps from the left limit $1/4$ to $1/2$.

7 The most singular phenomenon: close to one is not one

The endpoints $\theta = 0$ and $\theta = 1$ are reducible and must be treated separately.

Proposition 7.1 (Endpoint laws). *If $\theta = 0$, outcomes alternate deterministically after the first outcome. Hence*

$$\mathbb{P}_1(\tau_0 = \infty) = \frac{1}{2}, \quad \mathbb{P}_x(\tau_0 = \infty) = 1 \quad (x \geq 2).$$

If $\theta = 1$, the first outcome repeats forever, and

$$\mathbb{P}_x(\tau_0 = \infty) = \frac{1}{2} \quad (x \geq 1).$$

For every fixed $x \geq 1$,

$$\lim_{\theta \uparrow 1} \mathbb{P}_x(\tau_0 = \infty) = \frac{1}{4} \neq \frac{1}{2} = \mathbb{P}_x^{(\theta=1)}(\tau_0 = \infty). \quad (7.1)$$

Proof. The endpoint descriptions are immediate. For the left limit, use $r(\theta) \uparrow 1$ and $C(r) \rightarrow 3/4$ in (5.6). $\square$

The jump has a simple explanation. At $\theta = 1$, an initial win continues forever. If θ is only close to one, switches are rare, but they still occur infinitely often almost surely. Over an infinite time horizon, a rare switch is different from no switch.

Theorem 7.2 (Noncommuting target and persistence limits). *Let*

$$H_m(\theta) = \mathbb{P}_1(T_m < \tau_0), \quad m \geq 2.$$

Then

$$\lim_{\theta \uparrow 1} \lim_{m \rightarrow \infty} H_m(\theta) = \frac{1}{4}, \quad (7.2)$$

whereas

$$\lim_{m \rightarrow \infty} \lim_{\theta \uparrow 1} H_m(\theta) = \frac{1}{2}. \quad (7.3)$$

Proof. For fixed $\theta < 1$, Theorem 5.1 gives $\lim_m H_m(\theta) = 1 - C(r)$; Proposition 7.1 gives the first iterated limit. For fixed m , a first loss ruins the gambler. Conditional on a first win, the probability of retaining the winning state long enough to cross m tends to one as $\theta \uparrow 1$. Hence $\lim_{\theta \uparrow 1} H_m(\theta) = 1/2$, proving the second limit. $\square$

The jump can also be studied on a natural scale. The initial capital should be compared with the mean length of a run.

Theorem 7.3 (Persistence-capital transition window). *Let $\theta = 1 - \varepsilon$ with $\varepsilon \downarrow 0$, and let $x_\varepsilon \in \mathbb{Z}_{>0}$. If*

$$\varepsilon x_\varepsilon \longrightarrow \lambda \in [0, \infty],$$

then

$$\mathbb{P}_{x_\varepsilon}(\tau_0 = \infty) \longrightarrow 1 - \frac{3}{4}e^{-\lambda/2}. \quad (7.4)$$

If $\varepsilon x_\varepsilon \rightarrow \infty$, the limit is one.

Proof. Expansion of (5.4) at $\theta = 1$ gives

$$r(1 - \varepsilon) = 1 - \frac{\varepsilon}{2} + O(\varepsilon^2), \quad C(r) \longrightarrow \frac{3}{4}.$$

For finite λ , the expansion implies

$$r^{x_\varepsilon - 1} = \exp((x_\varepsilon - 1) \log r) \longrightarrow e^{-\lambda/2},$$

and (5.6) yields (7.4). Moreover, $\log r(1 - \varepsilon)/\varepsilon \rightarrow -1/2$; hence if $\varepsilon x_\varepsilon \rightarrow \infty$, then $(x_\varepsilon - 1) \log r \rightarrow -\infty$ and the ruin probability tends to zero. $\square$

The scale $x \asymp (1 - \theta)^{-1}$ is natural because it has the same order as the mean length of a persistent run. Capital much smaller than this scale cannot cover a typical losing run. With capital much larger than this scale, the ultimate survival probability is close to one.

Interpretation and scope

The result is not a method for turning a small bankroll into a fortune. It is a statement about an infinite-horizon first-passage probability. If

$$M = \sup_{0 \leq n < \tau_0} S_n$$

is allowed to take the value $+\infty$, then

$$\mathbb{P}_x(M = +\infty) = \mathbb{P}_x(\tau_0 = \infty).$$

The same surviving paths cross every finite target, so the corresponding hitting events are nested rather than disjoint. They do not form a probability mass function over target levels.

The model makes strong assumptions. The horizon is infinite, the rules never change, capital is integer-valued, and there are no costs, deadlines, or betting limits. Over a finite time, the relevant probability is $\mathbb{P}(T_m \leq N < \tau_0)$, which may be tiny for a huge m even when the infinite-horizon limit is positive. Proposition 3.2 puts a number on this point: in the fair stationary models, a surviving path takes about two rounds for each additional unit of target capital.

The Markov calculation shows that an average does not describe all the risk. The win rate and the expected payoff do not determine the ruin probability. Losses that come in runs can use up a small reserve before the positive drift has time to appear.

8 Conclusion

The original question seems to be about a distant target. In this model, it is mainly about early ruin. A path that survives forever eventually crosses every finite level. Starting from x units, the survival probability is $1 - \alpha^x$. This is why twenty units give a probability above 99.99% of crossing even a very large finite target, if there is no deadline.

The dependent model adds another point. Two games may have the same win rate and the same mean gain, but different ruin probabilities. It matters whether the losses are separated or come in runs. Near $\theta = 1$, “very unlikely to switch” is different from “unable to switch.” This difference appears as a discontinuity. The scale $x(1 - \theta)$ describes how much capital is needed to cover it.

In practice, a positive average is useful only if the player, firm, or strategy survives long enough to receive it. Capital also provides time. The casino in this example faces the same issue: a fair coin with payoff $+2/-1$ gives the player a positive expected gain. A real casino would not offer this game forever without limits.

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